

Harmonic Maps

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§ 1 Mikhail Karpukhin and Daniel Stern's Paper

(1.1) In this paper,

$$\Delta := dd^*,$$

which is a positive operator. Therefore, for harmonic map $u : M \rightarrow \mathbb{S}^k$, the E-L equation is

$$\Delta u = |du|^2 u.$$

Example 1.1 (Second fundamental form of a level set). Let $f : M \rightarrow \mathbb{R}$ be a smooth function, then $M_c := f^{-1}(c)$ is a hypersurface of M . Take $\nu = \text{grad } f$, then the second fundamental form is

$$\begin{aligned} A_\nu(X, Y) &= \langle \nu, \nabla_X Y \rangle \\ &= -\text{Hess}(f)(X, Y). \end{aligned}$$

In particular, for the sphere $\mathbb{S}^k(r)$, take $\nu = \frac{\partial}{\partial r}$, then

$$A_\nu(X, Y) = -r^{-1} \langle X, Y \rangle. \quad (1)$$

Example 1.2 (Hessian of distance in space form). Let ρ be a distance function on space form $M(K_0), g$, then

$$\text{Hess}(\rho) = \frac{f'(\rho)}{f(\rho)}(g - d\rho \otimes d\rho), \quad (2)$$

where

$$f(r) = \begin{cases} \sin(\sqrt{K_0}r)/\sqrt{K_0} & \text{if } K_0 > 0 \text{ and } r < \pi/\sqrt{K_0}, \\ r & \text{if } K_0 = 0, \\ \sinh(\sqrt{K_0}r)/\sqrt{K_0} & \text{if } K_0 < 0. \end{cases}$$

(1.2) For the system

$$\Delta u + \frac{DW(u)}{\epsilon^2} = 0,$$

since the nonlinear term does not involve the gradient of u , every weak solution should be **smooth** once we know DW is bounded. (Hint: we can use bootstrap argument.)

(1.3) For a function $u \in C^2(\Omega) \cap C(\bar{\Omega})$, we have **a priori estimates**

$$u(x) \leq \max_{\partial\Omega} |u| + C \max_{\Omega} |\Delta u| \quad (3)$$

for any $x \in \Omega$. (see [HL00], Proposition 2.15.)

(1.4) **Kato's type inequality:**

Let $V \rightarrow M$ be a vector bundle with a metric $\langle, \rangle$ and a compatible linear connection ∇ . Let $\xi \in \Gamma(V)$ be a smooth section. Then it's easy to get the classical Kato's inequality

$$|\nabla \xi| \geq |d|\xi||, \quad (4)$$

since, by Cauchy-Schwarz inequality,

$$2|\xi| \cdot (d|\xi|) = |d|\xi|^2| = 2|\langle \nabla \xi, \xi \rangle| \leq 2|\nabla \xi||\xi|.$$

Proposition 1.3 ([LW06]). Let $\phi : \mathbb{S}^m \rightarrow \mathbb{S}^k$ be a nonconstant smooth harmonic

map, then

$$|\nabla^2 \phi|^2 \geq \frac{m}{m-1} |\nabla |\nabla \phi||^2 \quad (5)$$

with equality holds at a point $x \in \mathbb{S}^m$ iff $\nabla^2 \phi(x) = 0$.

Remark 1.1. Note that

$$|\nabla |\nabla \phi||^2 = \frac{|\langle \nabla^2 \phi, \nabla \phi \rangle|^2}{|\nabla \phi|^2},$$

where

$$|\langle \nabla^2 \phi, \nabla \phi \rangle|^2 = \sum_{\alpha=1}^m \left[\sum_{i=1}^k \sum_{\beta=1}^m \phi_{\alpha\beta}^i \phi_{\beta}^i \right]^2.$$

However, the iff part of the proposition has some problem.(?) For example, take $k = m = 2$ and let

$$(\phi_{\alpha\beta}^1) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad (\phi_{\alpha\beta}^2) = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}, \quad (\phi_{\alpha}^1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad (\phi_{\alpha}^2) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

hold at a point x_0 , then the equality holds in (5), but $\nabla^2 \phi \neq 0$. Or consider the harmonic function $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $\phi(x, y) = x^2 - y^2$.

Remark 1.2. From the proof, it seems that the geometry of domain or target does not play a role, so it's still valid for general setting:

$$|\text{Hess } \phi|^2 \geq \frac{m}{m-1} |d|d\phi||^2$$

holds for any harmonic map $\phi : M^m \rightarrow N$.

(1.5) Mean value inequality

First of all, recall some facts:

Remark 1.3. 1. Change of Variables Formula: Suppose D and E are open domains of integration in $\mathbb{R}^n$, and $G : \bar{D} \rightarrow \bar{E}$ is smooth map that restricts to a diffeomorphism from D to E . For every continuous function $f : \bar{E} \rightarrow \mathbb{R}$,

$$\int_E f dV = \int_D (f \circ G) |\det DG| dV. \quad (6)$$

2. Laplacian acting on distance on space form: (Here $\Delta = -dd^*$)

$$\Delta r(x) = \begin{cases} (m-1)\sqrt{K} \cot(\sqrt{K}r) & \text{for } K > 0, \\ (m-1)r^{-1} & \text{for } K = 0, \\ (m-1)\sqrt{-K} \coth(\sqrt{-K}r) & \text{for } K < 0 \end{cases} \quad (7)$$

3. $\Delta r = H$, where $H(x) = \langle -\nabla_{e_i} e_i, \frac{\partial}{\partial r} \rangle$ is the mean value curvature of ∂B_r at x .

Suppose $f \in C^\infty(M^n)$, $B_s := B_s(x_0)$ is a geodesic ball and set $\Sigma_s := \partial B_s$. First, according to the first variation formula, we have

$$\begin{aligned} \frac{d}{ds} \int_{\Sigma_s} f dA_s &= \frac{d}{ds} \int_{\Sigma} f \circ \phi \cdot J(x, s) dA_s \\ &= \int_{\Sigma_s} \left(\frac{\partial f}{\partial \nu} + f \operatorname{div}_{\Sigma_s}(\nu) \right) \\ &= \int_{\Sigma_s} \left(\frac{\partial f}{\partial \nu} - f \Delta(d_{x_0}) \right). \end{aligned}$$

See [Li12]. Here we can set $\phi : \Sigma \times (-\epsilon, \epsilon) \rightarrow M$, and the variation field is the unit outer normal $\nu = d\phi(\frac{d}{ds})$.

$$\begin{aligned} &\frac{d}{ds} \left(s^{1-n} \int_{\partial B_s} f \right) \\ &= (1-n)s^{-n} \int_{\partial B_s} f + s^{1-n} \int_{\partial B_s} \left(\frac{\partial f}{\partial \nu} - f \Delta(d_{x_0}) \right) \\ &= (1-n)s^{-n} \int_{\partial B_s} f - s^{1-n} \int_{B_s} \Delta f - \int_{\partial B_s} f \Delta(d_{x_0}) \\ &= -s^{1-n} \int_{B_s} \Delta f + s^{1-n} \int_{\partial B_s} f \cdot \left(\frac{(1-n)}{d_{x_0}} - \Delta d_{x_0} \right). \end{aligned}$$

By **Hessian** comparison theorem, we have

$$\frac{(1-n)}{d_{x_0}} - \Delta d_{x_0} \geq -C(n, K),$$

where K is the sectional curvature bound for M . Therefore,

$$\frac{d}{ds} \left(s^{1-n} \int_{\partial B_s} f \right) \geq -s^{1-n} \int_{B_s} \Delta f - C(n, K) s^{1-n} \int_{\partial B_s} f. \quad (8)$$

(1.6) Universal lower bound on the energy

Proposition 1.4. *For any closed Riemannian manifold (M, g) , there exists positive constant $\beta(M)$, s.t.*

$$E(u) \geq \beta \tag{9}$$

for any nonconstant harmonic map $u : M \rightarrow \mathbb{S}^k$ of any dimension $k \in \mathbb{N}$.

The gradients of the proof includes monotonicity formula, Bochner formula and small energy regularity.

(1.7) Rayleigh principle

Reference: [Eva10, §6.5, §6.6, Appendix D], [Spectral analysis](#), by F. Haslinger and reference therein, [Hel13, chap 11], [Ros97], [Cha84], [Tho79, P84-86].

In this paper, Karpukhin and Stern investigate the following variation form

$$\nu_m := \inf_{F_m} \sup_{u \in F_m, ||u||=1} \langle Lu, u \rangle, \tag{10}$$

which is different from the usual form

$$\lambda_m = \sup_{F_m} \inf_{u \in (F_m)^\perp, ||u||=1} \langle Lu, u \rangle, \tag{11}$$

or

$$\lambda_m = \inf_{u \in (K_m)^\perp, ||u||=1} \langle Lu, u \rangle, \quad K_m := \text{span}\{\lambda_1, \dots, \lambda_m\}. \tag{12}$$

However, ν should coincide with λ_m : Take $F_m = K_m$, the span of first m eigenfunctions $\{v_1, \dots, v_m\}$. Writing $u = \sum_{i=1}^m u^i v_i$, then (since $||u|| = 1$)

$$\langle Lu, u \rangle = \sum_{i=1}^m (u^i)^2 \lambda_i \leq \lambda_m,$$

which implies $\sup_{K_m} \langle Lu, u \rangle = \lambda_m$. Now it's easy to see this is the minima of (10).

§ 2 Lin-Wang, the analysis of harmonic maps and their heat flow

(2.1) (Minimizing harmonic maps in higher dimensions may have singularities.)

Proposition 2.1. For $n \geq 3$, $\phi_0(x) = \frac{x}{|x|} : \mathbb{B}^n \rightarrow \mathbb{S}^{n-1}$ is a minimizing harmonic map.

Since

$$\phi^j = \frac{x_j}{|x|}, \quad \phi_i^j = \frac{\delta_{ij}}{|x|} - \frac{x_j x_i}{|x|^3}, \quad \phi_{ik}^j = -\frac{\delta_{ij} x_k}{|x|^3} - \frac{\delta_{jk} x_i + \delta_{ik} x_j}{|x|^3} + 3 \frac{x_i x_j x_k}{|x|^5},$$

thus

$$|d\phi|^2 = \sum_{i,j} \left(\frac{\delta_{ij}}{|x|} - \frac{x_j x_i}{|x|^3} \right)^2 = \sum_j \left(\frac{1}{|x|^2} + \frac{x_j^2}{|x|^4} - 2 \frac{x_j^2}{|x|^4} \right) = \frac{n}{|x|^2} - \frac{1}{|x|^2}$$

$$\Delta \phi^j = -\frac{x_j}{|x|^3} - \frac{x_j + nx_j}{|x|^3} + 3 \frac{x_j}{|x|^3} = (1-n) \frac{x_j}{|x|^3} = -|d\phi|^2 \phi^j(x).$$

And

$$\operatorname{div}(\langle \nabla \phi, \phi \rangle) = \langle \nabla_{e_i} (\nabla \phi) \phi, e_i \rangle = \phi_i^j \phi_j^i + \phi_{ji}^i \phi^j$$

$$\operatorname{div}(\operatorname{div}(\phi) \phi) = \operatorname{div}(\phi_i^i \phi^j e_j) = \phi_i^i \phi_j^j + \phi_{ij}^i \phi^j$$

since

$$\nabla \phi^j = \sum_i \phi_i^j e_i, \quad (\nabla \phi) \phi = \nabla_\phi \phi = \phi^j \nabla_{e_j} (\phi^k e_k) = \phi^j \phi_j^k e_k$$

And

$$\sum_{i \neq j} a_{ij} a_{ji} \geq - \sum_i \left(\sum_j (a_{ij})^2 \right)^{\frac{1}{2}} \left(\sum_j (a_{ji})^2 \right)^{\frac{1}{2}} \geq - \left(\sum_i \left(\sum_j (a_{ij})^2 \right) \right)^{\frac{1}{2}} \left(\sum_i \left(\sum_j (a_{ji})^2 \right) \right)^{\frac{1}{2}}.$$

(2.2) (Optimal regularity) A slight modification of this example shows that a minimizing harmonic map from $\mathbb{R}^n, n \geq 3$, can have $(n-3)$ -dimensional singular set. In fact, $\phi(x, y) = \frac{x}{|x|} : \mathbb{R}^3 \times \mathbb{R}^{n-3} \rightarrow \mathbb{S}^2$ is a minimizing harmonic map whose singular set is $\{0\} \times \mathbb{R}^{n-3}$. In general, Schoen and Uhlenbeck have shown in their pioneering work that Hausdorff dimension of the singular set of any minimizing harmonic map is of at most $(n-3)$ -dimension.

(2.3) (Iteration) On Lemma 2.2.7, P18: The lemma states that, for some $\epsilon > 0, \theta_0 \in$

$(0, \frac{1}{2})$,

$$E(u, B_1) \leq \epsilon_0^2 \Rightarrow E(u_{\theta_0}, B_1) \leq \frac{1}{2} E(u, B_1). \quad (13)$$

Here we used

$$r^{2-n} E(u, B_r) = E(u_r, 1), \quad u_r(x) := u(rx). \quad (14)$$

For,

$$\int_{B_1} |\nabla u_r|^2(x) dx = \int_{B_1} |\nabla u|^2(rx) r^2 dx = r^{2-n} \int_{B_r} |\nabla u|^2(y) dy.$$

Lemma 2.2. Suppose

$$\phi(\rho r) \leq \theta \phi(r), \quad \text{for some fixed } \rho \in (0, 1), \theta > 0.$$

and $\phi(r) < \infty$. Then

$$\phi(s) \leq C s^\alpha, \quad s \in (0, r].$$

Here $\alpha = \frac{\log \theta}{\log \rho}$ (so if $\theta < 1 \Rightarrow \alpha > 0$), and $C = r^{-\alpha} \phi(r)$.

Proof. By iterating this estimate, we get

$$\phi(\rho^k r) \leq \theta^k \phi(r), \quad \text{for any } k \in \mathbb{N}.$$

Now, for any $s \in (0, r]$, we can write $s = \rho^k r$, so that

$$k = \frac{\log(s/r)}{\log \rho} = \log_\rho(s/r).$$

Then,

$$\phi(s) = \phi(\rho^k r) \leq \theta^k \phi(r) = \phi(r) \cdot (\theta^{\log_{1/\rho}(r/s)}) = \phi(r) \cdot \left((\theta^{1/\log(1/\rho)})^{\log(r/s)} \right)$$

Define

$$\alpha := \frac{\log \theta}{\log \rho}, \quad (15)$$

then

$$\theta^k = (\rho^k)^\alpha = \left(\frac{s}{r}\right)^\alpha.$$

Thus,

$$\phi(s) \leq \theta^{\log_\rho(s/r)} \phi(r) = \phi(r) \left(\frac{s}{r}\right)^\alpha = C s^\alpha.$$

□

Go back to the book, Page 21, we know

$$r_0^{2-n} E(u, B_{r_0}) = E(u_{r_0}, B_1) \leq \epsilon_0^2.$$

From (13), we have

$$E(u_{\theta_0 r_0}, B_1) \leq \frac{1}{2} E(u_{r_0}, B_1) \leq \frac{1}{2} \epsilon_0^2.$$

Using (13) again,

$$E(u_{\theta_0^2 r_0}, B_1) \leq \frac{1}{4} E(u_{r_0}, B_1) \leq \frac{1}{4} \epsilon_0^2.$$

Repeat this process, we have

$$E(u_{\theta_0^k r_0}, B_1) \leq \left(\frac{1}{2}\right)^k E(u_{r_0}, B_1).$$

Letting $\rho = \theta_0, \theta = \frac{1}{2}$ in the above lemma, we get

$$E(u_r, B_1) \leq C r^\alpha.$$

§ 3 Lin-Wang, Harmonic and quasi-harmonic spheres, Part I

(3.1) (Density of measure)

(3.2) Why

$$\frac{d}{dr}(r^{-m} v(B_R)) = 0 \Rightarrow v(B_R) = 0?$$

From small regularity property, we have

(3.3) (The singular set Σ is closed and locally finite) Suppose Σ has infinite many points, then

$$\liminf_{\epsilon \rightarrow 0} \int_M |\nabla u_\epsilon|^2 \geq \sum \epsilon_0^2 = \infty.$$

(3.4) (Similar derivation of the Pohozaev identity)

$$-\Delta v + \frac{1}{c^2} f(v) = 0, \quad \text{in } \mathbb{R}^2.$$

Multiplying it with $\phi_n(x)x \cdot Dv$ and integrate them over $\mathbb{R}^2$,

$$\int_{\mathbb{R}^2} \Delta v \phi_n x Dv + \frac{1}{c^2} DF(v) \phi_n x Dv dx = 0.$$

That is

$$\int_{\mathbb{R}^2} -v_{ii}^\alpha v_k^\alpha x^k \phi_n + \frac{1}{c^2} F_\alpha v_k^\alpha x^k \phi_n dx = 0.$$

Since

$$\begin{aligned} \int_{\mathbb{R}^2} F_\alpha v_k^\alpha x^k \phi_n dx &= \int_{\mathbb{R}^2} (F(v))_k x^k \phi_n dx = - \int_{\mathbb{R}^2} F(v) x^k D_k \phi_n dx - 2 \int_{\mathbb{R}^2} F(v) \phi_n \\ &\quad - \int_{\mathbb{R}^2} v_{ii}^\alpha v_k^\alpha x^k \phi_n = \int_{B_{2n}} v_i^\alpha v_{ki}^\alpha x^k \phi_n + \int_{B_{2n}-B_n} v_i^\alpha v_k^\alpha x^k D_i \phi_n + \int_{\mathbb{R}^2} (v_i^\alpha)^2 \phi_n \\ &\leq \frac{1}{2} \int_{B_{2n}} (v_i^\alpha v_i^\alpha)_k x^k \phi_n + C \int_{B_{2n}-B_n} e(v) + \int_{\mathbb{R}^2} (v_i^\alpha)^2 \phi_n \\ &\leq C \int_{B_{2n} \setminus B_n} e(v) \end{aligned}$$

Thus,

$$\begin{aligned} \frac{2}{c^2} \int_{\mathbb{R}^2} F(v) \phi_n &= - \frac{2}{c^2} \int_{B_{2n} \setminus B_n} F(v) x^k D_k \phi_n dx + \int_{B_{2n}} v_i^\alpha v_{ki}^\alpha x^k \phi_n + \int_{B_{2n}-B_n} v_i^\alpha v_k^\alpha x^k D_i \phi_n + \int_{\mathbb{R}^2} (v_i^\alpha)^2 \phi_n \\ &\leq \frac{2\epsilon}{c^2} \int_{B_{2n} \setminus B_n} F(v) x^k D_k \phi_n dx + C_2 \int_{B_{2n} \setminus B_n} e(v). \end{aligned}$$

Note that $e(v)$ in the paper includes $F(v)$ term!

(3.5) (On Hölder spaces) Recall the Hölder norm:

$$\|u\|_{C^{k,\gamma}(\bar{U})} := \sum_{|\alpha| \leq k} \|D^\alpha u\|_{C(\bar{U})} + \sum_{|\alpha|=k} [D^\alpha u]_{C^{0,\gamma}(\bar{U})}. \quad (16)$$

There is a domain Ω such that $C^1(\bar{\Omega}) \not\subset C^\alpha(\bar{\Omega})$, see Gilbarg-Trudinger's book, P53. However, the mean value inequality

$$|f(x) - f(y)| \leq \max |Df| |x - y|$$

holds for **convex** domain.

First, one has the following compactness fact:

Proposition 3.1 (cf. Corollary 11.38 of Giaquinta-Martinazzi's book). *The immersion $C^{r,\alpha}(\bar{\Omega}) \rightarrow C^r(\bar{\Omega})$, for $\alpha \in (0, 1]$, $r \in \mathbb{N}$ is compact.*

More generally:

Lemma 3.2 (cf. P136 of Gilbarg-Trudinger's book). *Let Ω be a $C^{k,\alpha}$ domain in $\mathbb{R}^n$ (with $k \geq 1$) and let S be a bounded set in $C^{k,\alpha}(\bar{\Omega})$. Then S is precompact in $C^{j,\beta}(\bar{\Omega})$ if $j + \beta < k + \alpha$.*

(3.6) P404, $u_\epsilon \rightarrow u_*$ in $C^1(B_1 \setminus \Sigma) \cap H^1(B_1 \setminus \Sigma)$: By iteration (2.3), we see that $\{u_\epsilon\}$ has a uniform C^α semi-norm bound, which implies the equicontinuity. We need the uniform bound of $\{\sup |u_\epsilon|\}$ to apply Arzelà-Ascoli theorem (or the Lemma above), which is promised by the assumption that N is compact!

§ 4 f-harmonic maps

Definition 4.1 (f -Laplacian, Ric_f).

$$\Delta_f(u) := \Delta u - \langle \nabla u, \nabla f \rangle,$$

$$\text{Ric}_f := \text{Ric} + \text{Hess } f.$$

Proposition 4.2 (Bochner formula).

$$\frac{1}{2} \Delta_f |\nabla u|^2 = |\text{Hess } u|^2 + \langle \nabla u, \nabla \Delta_f u \rangle + \text{Ric}_f(\nabla u, \nabla u). \quad (17)$$

Indeed,

$$\begin{aligned} \frac{1}{2} \Delta_f |\nabla u|^2 &= \frac{1}{2} \Delta |\nabla u|^2 - \text{Hess } u(\nabla u, \nabla f) \\ \langle \nabla u, \nabla \Delta_f u \rangle &= \langle \nabla u, \nabla \Delta u \rangle - \text{Hess } u(\nabla u, \nabla f) - \text{Hess } f(\nabla u, \nabla u). \end{aligned}$$

For the first inequality, recall

$$\text{Hess } u(X, Y) = \langle \nabla_X \nabla u, Y \rangle.$$

The second equality follows from

$$\begin{aligned}\langle \nabla u, \nabla \langle \nabla u, \nabla f \rangle \rangle &= \nabla u \cdot \langle \nabla u, \nabla f \rangle \\ &= \nabla u \cdot \nabla u \cdot f = \nabla u \cdot \nabla f \cdot u\end{aligned}$$

and

$$\begin{aligned}\text{Hess } u(\nabla u, \nabla f) + \text{Hess } f(\nabla u, \nabla u) &= \nabla u \cdot \nabla f \cdot u - (\nabla_{\nabla u} \nabla f)u + \nabla u \cdot \nabla u \cdot f - (\nabla_{\nabla u} \nabla u)f \\ &= \nabla u \cdot \nabla f \cdot u + \nabla u \cdot \nabla u \cdot f - (\nabla_{\nabla u} \nabla f)u - (\nabla_{\nabla u} \nabla u)f \\ &= 2\langle \nabla u, \nabla \langle \nabla u, \nabla f \rangle \rangle - \nabla u \cdot \langle \nabla u, \nabla f \rangle.\end{aligned}$$

§ 5 f-pseudoharmonic maps

$$E_{b,f}(u) = \frac{1}{2} \int_M |d_b u|^2 e^{-f} dv_\theta \quad (18)$$

(5.1) (The first variation formula.)

Let $u_t : M \rightarrow N$ be a family maps from pseudohermitian manifold (M, H, θ) to Riemannian manifold N . Then

$$\begin{aligned}\frac{d}{dt} E_{b,f}(u_t) &= \int_M \langle \nabla_{\frac{\partial}{\partial t}} du_t(e_i), du_t(e_i) \rangle e^{-f} dv_\theta \\ &= \int_M \langle \nabla_{e_i} \frac{\partial u_t}{\partial t}, du_t(e_i) \rangle e^{-f} dv_\theta \\ &= \int_M e_i \langle V, e^{-f} du(e_i) \rangle - \langle V, \nabla_{e_i} (e^{-f} du(e_i)) \rangle dv_\theta \\ &= \int_M e_i \langle V, e^{-f} du(e_i) \rangle dv_\theta + \int_M \langle V, e_i(f) du(e_i) - \nabla_{e_i} du(e_i) \rangle e^{-f} dv_\theta.\end{aligned}$$

Set 1-form $\eta(X) := \langle V, e^{-f} du(X) \rangle$, then

$$\begin{aligned}\delta(\eta) &= - \sum_{A=1}^{2m+1} (\nabla_{e_A} \eta)(e_A) \\ &= - e_A \langle V, e^{-f} du(e_A) \rangle + \langle V, e^{-f} du(\nabla_{e_A} e_A) \rangle.\end{aligned}$$

It follows that

$$\begin{aligned} \frac{d}{dt} E_{b,f}(u_t) &= - \int_M \delta(\eta) + T \langle V, e^{-f} du(T) \rangle - \langle V, e^{-f} du(\nabla_T T) \rangle dv_\theta \\ &\quad - \int_M \langle V, \tau_b(u) \rangle e^{-f} dv_\theta + \int_M \langle V, du(\nabla_b f) \rangle e^{-f} dv_\theta \\ &= - \int_M \langle V, \tau_b(u) - du(\nabla_b f) \rangle e^{-f} dv_\theta. \end{aligned}$$

Thus u is f -pseudoharmonic iff

$$\tau_b(u) - du(\nabla_b f) = 0, \quad (19)$$

or, locally,

$$u_{ii}^\alpha - f_i u_i^\alpha = 0, \quad \forall \alpha. \quad (20)$$

(5.2) (Bochner formula)

Note that $|d_b u|^2 = \sum_{i,\alpha} (u_i^\alpha)^2$

$$\begin{aligned} &\Delta_b |d_b u|^2 - \langle d |d_b u|^2, d_b f \rangle \\ &= \Delta_b |d_b u|^2 - 2u_{ij}^\alpha u_i^\alpha f_j \end{aligned}$$

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